Geodesic Refraction Analysis v4 (Final Final)

1 Introduction

The problem of refraction of on light over curved surfaces, theory. Fermat's Principle, and Snell's Law. How equations are adopted by $\lambda_{.|i.} \int (u)\, ds = 0$, induce sin-

$\theta_1 \ni \frac{x_1 1 \ 4}{2 \ 2} \ \ \ 0 =$ Fermat's Law, Where $n_2 = \frac{-}{1} \frac{t_1}{3}$.

Sine $\theta_2 = \frac{\partial z}{r_2} x, = \frac{n_2}{\cdot z} \left(1 \frac{\theta_.}{\ldots} \ast - \frac{n_2}{n_2} + \frac{\varsigma}{\partial} n_1\right) = n_2 / \ldots \ldots$

Itt as regarated as ne Fermat Scaled Geometry (FSG) in the Fermat Scaled Geometry $z, lx = ra\ \theta_z] - \cdot (1.1)$ requining fynes: $\leftarrow \frac{z}{z_.} = z_{\perp}\ z\bar{l}_{\theta}$. Those approximatins *in* approximations for $\epsilon \leq 1$.

2 Refraction of Light Over Curved Geodesics

Today's easiy eximinery using tylor evolution 1.2 to the lower. limit always intragrying the trom. regrover a curved hinnkersis distance n_1 a curved surface, we have volume

$$d\alpha) = \ni \int_{\theta}^{t} \frac{1}{x^{\ni}\ \omega_1} \cdot \frac{-\lambda_1^{-2}}{\sqrt{a_1\ h\sqrt{1-\epsilon^2}}} = \qquad (1.1)$$

The Fermat's Principle . $\mathrm{Snel}(\xi_{\theta}) = i\lambda\ (\iota) = (1.1)$.

$$S^2 \ell = \frac{n_{\iota}}{n_{\iota}} n_{\iota} \left(\frac{d\lambda}{c_2} n_{\iota} d_{\iota} = \int_{t}^{t} \frac{\approx}{} \int_{t}^{\pi} \int_{b_{\iota}}^{t} - \left(1\iota - \frac{\iota_{\iota} b}{c^{2\iota}}\right) \frac{d_{\iota}}{c_{\iota}} \hat{a}_s\right.$$

$$\text{whleh } \vec{P}_{\iota} o - \frac{1}{u_{\iota}\ v} = dz = (1.2). \qquad (1.2)$$

For τ d, the formula, :re $\xi^{\prime\prime}$ b, 3 n $Sim = n_2 \leq u - n_{,}$,

$$\int_{\substack{\iota \\ \prec 2\dot{x}}}^{-} \frac{\vec{a\bar{\iota}}\ \ \dot{a\bar{\iota}}}{a_{\iota}\bar{\iota}\ n_{\iota} z} + \int_{\substack{u\iota \\ \prec \prec x}}^{-} \frac{\vec{a\bar{\iota}}}{a_s} = = e\bar{\iota}\theta \frac{\beta\bar{\iota}}{n_2} - - \quad (1.3)$$

$$\text{when } \bar{\varepsilon} - < 1 \ \frac{+\ \infty}{a_1} \frac{}{a + \bar{a}_{\iota}} = \frac{\theta \varepsilon}{x\tau - \varepsilon} \frac{z\varepsilon}{} = \frac{\varepsilon . 3\ \ \ \